

8.FOURIER TRANSFORM AND FREQUENCY DOMAIN ANALYSIS

Experiment 8.1% Fourier Transform of Periodic Impulse Train

```
clc;

clear;

close all;

% Time axis

t = -5:0.01:5;

% Generate periodic impulse train

x = zeros(size(t));

x(1:100:end) = 1; % Impulse every 100 samples

% Fourier Transform

X = fftshift(fft(x));

% Frequency axis

N = length(X);

f = linspace(-50,50,N);

% Plotting

figure;

subplot(2,1,1);

stem(t,x,'filled');

grid on;

title('Periodic Impulse Train');

xlabel('Time');

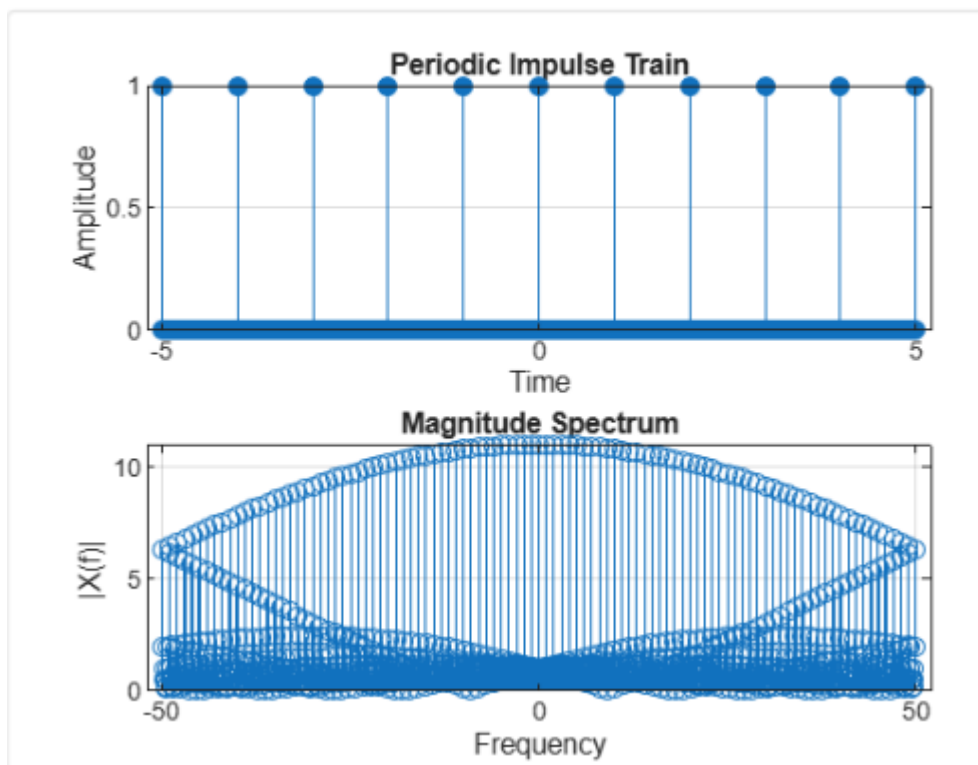
ylabel('Amplitude');

subplot(2,1,2);
```

```

stem(f,abs(X));
grid on;
title('Magnitude Spectrum');
xlabel('Frequency');
ylabel('|X(f)|');

```



Experiment 8.2% Discrete-Time Fourier Transform (DTFT)

```

clc;
clear;
close all;
% Input sequence
x = [1 2 3 4];
% Frequency range
w = -pi:0.01:pi;

```

% DTFT computation

X = zeros(size(w));

for k = 1:length(w)

for n = 1:length(x)

X(k) = X(k) + x(n)*exp(-1j*w(k)*(n-1));

end

end

% Plotting

figure;

subplot(2,1,1);

stem(0:length(x)-1,x,'filled');

grid on;

title('Input Sequence x(n)');

xlabel('n');

ylabel('Amplitude');

subplot(2,1,2);

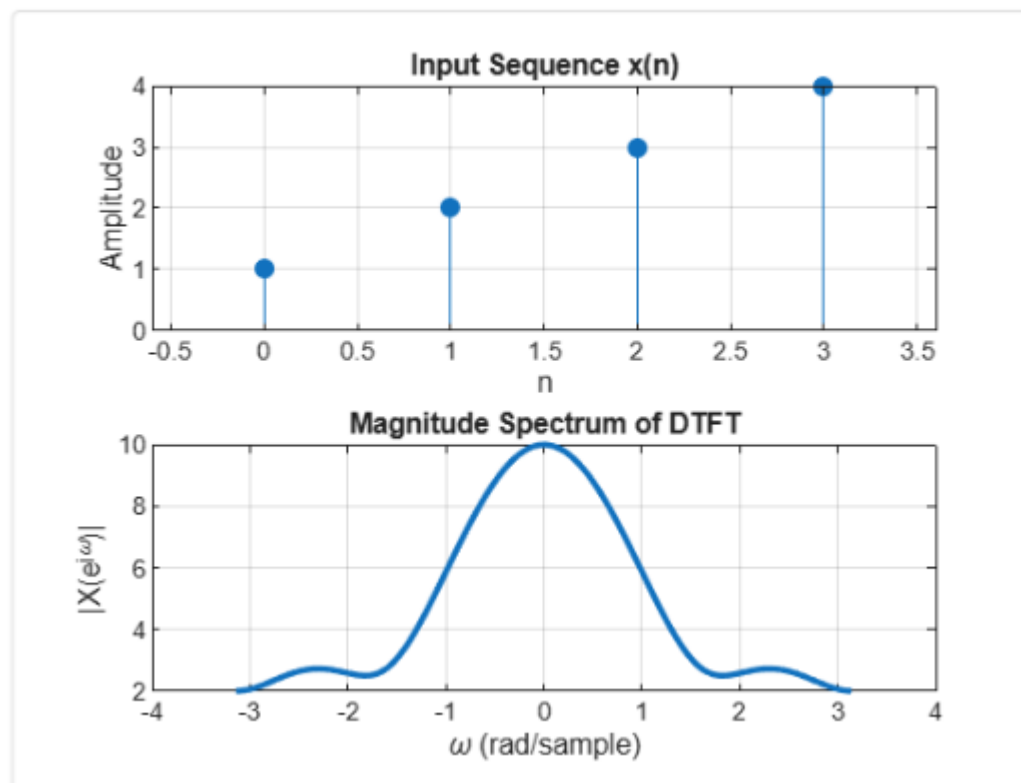
plot(w,abs(X),'LineWidth',2);

grid on;

title('Magnitude Spectrum of DTFT');

xlabel('\omega (rad/sample)');

```
ylabel('|X(e^{j\omega})|');
```



Experiment 8.3

% DTFT of a Rectangular Pulse

```
clc;
```

```
clear;
```

```
close all;
```

% Rectangular pulse

```
N = 10;
```

```
x = ones(1,N);
```

% Frequency range

```
w = -pi:0.01:pi;
```

% DTFT computation

```
X = zeros(size(w));
```

```
for k = 1:length(w)
```

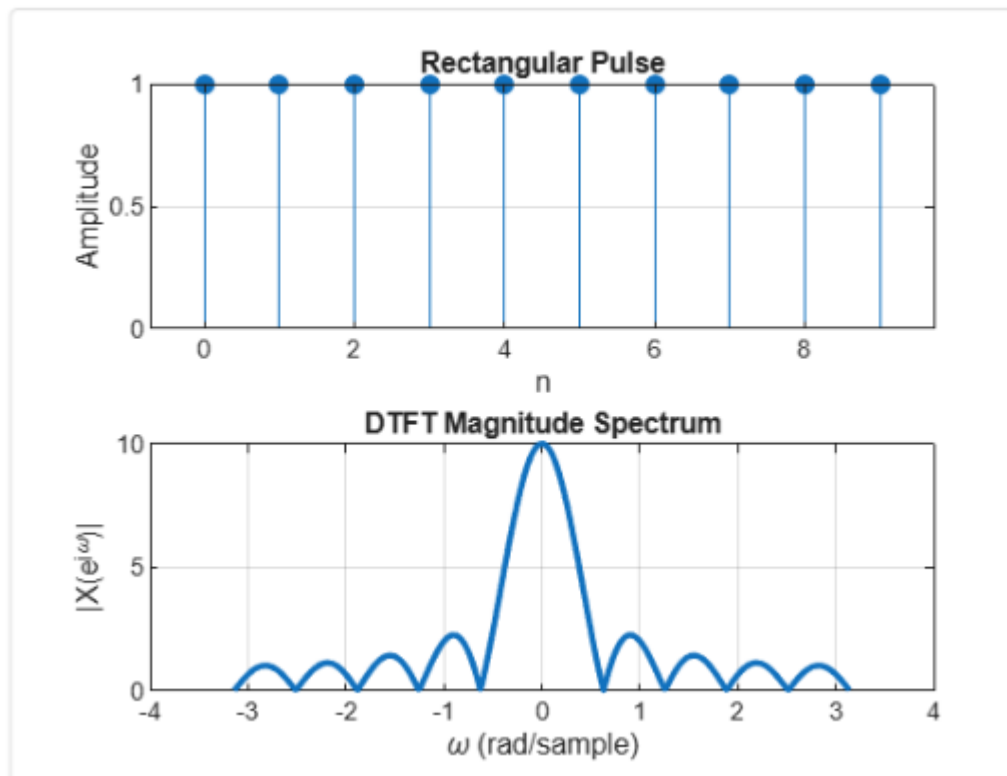
```
for n = 1:N
```

```
X(k) = X(k) + x(n)*exp(-1j*w(k)*(n-1));
```

```

end
end
% Plotting
figure;
subplot(2,1,1);
stem(0:N-1,x,'filled');
grid on;
title('Rectangular Pulse');
xlabel('n');
ylabel('Amplitude');
subplot(2,1,2);
plot(w,abs(X),'LineWidth',2);
grid on;
title('DTFT Magnitude Spectrum');
xlabel('\omega (rad/sample)');
ylabel('|X(e^{j\omega})|');

```



Experiment 8.4 % Realization of DIT-FFT for Spectral Analysis

```
clc;
clear;
close all;
% Input sequence
x = [1 2 3 4 5 6 7 8];
% FFT computation
X = fft(x);
% Number of samples
N = length(x);
% Frequency axis
f = 0:N-1;
% Display FFT values
disp('FFT Output:');
disp(X);
% Plotting
figure;
subplot(2,1,1);
stem(0:N-1,x,'filled');
grid on;
title('Input Sequence');
xlabel('n');
ylabel('Amplitude');
subplot(2,1,2);
stem(f,abs(X),'filled');
```

```
grid on;
```

```
title('Magnitude Spectrum using DIT-FFT');
```

```
xlabel('Frequency Index (k)');
```

```
ylabel('|X(k)|');
```

FFT Output:

```
36.0000 + 0.0000i -4.0000 + 9.6569i -4.0000 + 4.0000i -4.0000 +
```

